

TRADING LAB PRODUCT BLUEPRINT

A phased execution specification for an evidence-driven trading research, strategy testing, decision and journaling system.

Version 1.0
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Core product decision

Build a clean Trading Lab module inside the existing Buddies OS. Reuse authentication, design, deployment and historical data; replace the old analysis architecture only after the new Lab is verified.

Scope: Gold-first research, general strategy testing, AI-assisted rule authoring, three-pillar live decisions, manual execution, trade journal, and a reusable ladder campaign.

1. Executive decisions

This document records the product choices agreed before implementation begins.

Decision	Agreed direction
Build approach	Create a new Trading Lab module, migrate useful data/features, and retire obsolete Trading Agent code only after validation.
Execution boundary	Buddies OS provides research, testing, decisions and journaling. Trades are executed manually through an external broker.
Initial market	Gold first: XAU/USD for spot context and COMEX gold futures for authoritative exchange volume.
Strategy scope	General short-term and long-term strategy builder. Swing and controlled doubling are templates, not hardcoded limits.
AI role	AI converts ideas into explicit rules and explains deterministic results; it never manufactures data or backtest outcomes.
Ladder	Preserve and expand the ladder as a reusable campaign/risk overlay attachable to any strategy.
Connectors	User-manageable connector settings with encrypted server-side secret storage and provider capability routing.

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2. Migration from the current Trading Agent

The old Trading Agent remains available while the new module is built. The migration is selective: preserve proven assets and data, replace architectural weaknesses, then switch navigation after production acceptance.

Reuse	Replace
Buddies OS authentication and navigation	Separate and inconsistent Trading AI endpoints
Supabase project and existing user records	Browser-supplied decision evidence
AI provider configuration	Hardcoded models and silent failure handling
Watchlists and symbol preferences	One-timeframe technical logic
Existing trades, summaries and ladder progress	Missing-volume-as-low-volume behavior
Design system and useful chart components	Broker-execution emphasis and hardcoded strategies
Production deployment and CI foundation	Drifted trading schema and loosely linked records

No destructive cutover

Historical records stay intact. The old UI is removed only after migrated records, permissions, calculations and production behavior are verified.

Migration states

- Existing Trading Agent remains the current interface.
- Trading Lab is developed behind a protected route or feature flag.
- Existing trades and ladder campaigns are mapped to the new schema.
- Acceptance testing compares both systems.
- Trading navigation points to the new Lab.
- Old code is archived briefly, then removed after a stable production period.

3. Target architecture

The Trading Lab separates presentation, data, rules, AI and personal records. This prevents an AI response or one provider failure from becoming an unverified trading signal.

Layer	Responsibilities	Must not do
Connector Hub	Credentials, health, capability routing, fallbacks	Expose secrets to the browser
Market Data Hub	Normalize prices, volume, macro, news and timestamps	Treat missing data as a signal
Strategy Engine	Versioned rules, filters, exits, sizing and safety limits	Change rules during a test
Backtest Engine	Execute rules over historical data and save every fill	Use future information
AI Copilot	Draft rules, identify ambiguity, explain verified evidence	Invent trades or results
Decision Desk	Evaluate fundamental, technical and volume pillars	Override deterministic safety gates
Journal	Link decisions, plans, executions, psychology and lessons	Alter historical snapshots
Ladder	Apply progression rules to strategies or paper campaigns	Permit unlimited risk escalation

System flow

CONNECTORS -> NORMALIZED MARKET DATA -> STRATEGY / DECISION ENGINES
-> VERIFIED RESULTS -> AI EXPLANATION -> MANUAL ACTION
-> JOURNAL + LADDER PROGRESS -> REVIEW AND IMPROVEMENT

Fail-safe principle

When a required source is stale, unavailable or contradictory, confidence falls and the system moves toward NO TRADE. It never fills the gap with AI-generated facts.

4. Connector Hub and secret security

The user can add providers from a friendly settings page. The key passes once over HTTPS to authenticated server code, is stored encrypted, and is never returned to the browser in full.

Frontend action	Server behavior
Add API key	Authenticate owner, validate format, encrypt and store secret, save metadata reference
Test connection	Make a minimal provider request and record latency/capabilities
View connector	Return provider, masked suffix, status, last success and last error - never plaintext
Rotate key	Create new encrypted secret, verify it, then retire the old secret
Disable	Stop routing requests without deleting configuration
Delete	Remove encrypted secret and metadata after explicit confirmation

Security requirements

- Secrets never enter browser storage, URLs, public JavaScript, analytics, error bodies or normal logs.
- Supabase Vault or an equivalent server-side secret manager stores encrypted credentials; browser-accessible tables contain only metadata and secret references.
- Every connector action checks authenticated ownership and is recorded in a security audit trail.
- Row Level Security protects connector metadata; decrypted secrets are unavailable through the public Data API.
- Shared application credentials and personal connector credentials remain separate.
- Legacy Supabase anon/service-role usage is reviewed for migration to publishable/secret keys.

Capability routing

Instead of binding the product to one vendor, each connector declares capabilities such as live quote, historical bars, futures volume, open interest, trades, market depth, macro series, news, AI or webhook events. A user can select a primary and fallback provider per capability.

5. Market-data and charting plan

Need	Initial provider	Quality rule
Spot XAU/USD context	Twelve Data	Label as spot/provider quote and show timestamp
COMEX gold historical data	Databento GLBX.MDP3	Use exchange-derived futures records
Volume and open interest	Databento/CME statistics	Never substitute a missing field with zero
Trades and aggressor flow	Databento trades/MBP	Declare schema and contract
Deep order book	Databento MBO/MBP; Bookmap externally	Later phase due processing and licensing
Macro/yields	FRED	Show observation date and update age
Positioning	CFTC COT	Weekly context only, never intraday trigger
News/calendar	Dedicated connector	Source, event time and impact classification
Charts	TradingView Lightweight Charts	Our data feed; chart library is not a data provider
External alerts	TradingView webhook	Authenticated, deduplicated event ingestion

Volume integrity

Gold volume conclusions should be based on COMEX futures data. Spot XAU/USD volume may be provider-specific tick volume and must be labelled as a proxy if used at all.

Every data payload carries provider, symbol/contract, market, timeframe, as-of time, ingestion time, delay class, freshness state and quality state. Futures logic also records the active contract and roll method.

6. General Strategy Lab

Swing and doubling are starting templates. The underlying builder must support any measurable short-term or long-term strategy.

Strategy section	Configurable fields
Universe	Symbols, market, long/short permissions, date range
Timeframes	Context, setup and trigger intervals
Entry	Price, structure, indicator, volume, Wyckoff, macro and session conditions
Logic	ALL/ANY groups, nested conditions, cross-timeframe references
Exit	Fixed/ATR/structure stop, targets, trailing, partial and time exit
Sizing	Fixed, dollar risk, percent risk, volatility adjusted, progressive
Safety	Daily loss, drawdown, concurrent positions, cooldown and event veto
Execution	Bar close/next open, spread, fees, slippage and fill assumptions
Governance	Version, author, status, notes and immutable test snapshot

Initial templates

- Gold swing: D1/H4 regime, H1 pullback and confirmation, structure-based stop, 2R or next major resistance.
- Controlled doubling: a risk overlay with capped multiplier, maximum increases, margin checks and drawdown stop.
- Reversal/VSA: extreme location, climatic effort, rejection, confirmation and clean reward-to-risk.
- Momentum/pullback: break, acceptance, high-effort expansion, low-effort pullback and trigger.
- Blank strategy: completely configurable rule set.

Key distinction

A doubling or ladder model controls capital progression. It does not create a market edge by itself and must always be attached to a defined entry/exit strategy.

7. Ladder as a reusable campaign

The ladder becomes a first-class experiment that can be attached to any strategy, backtest, paper campaign or manually logged external trade sequence.

Configuration	Examples
Capital	Starting balance, step targets, withdrawal points
Progression	Advance, remain, step back or reset rules
Risk	Risk per step, multiplier, maximum size and maximum doublings
Activity	Maximum trades per step/session, cooldown and expiry
Failure	Step loss limit, campaign drawdown stop, restart behavior
Attachment	Strategy version, decision snapshots and journal trades
Mode	Historical simulation, live paper campaign or external-trade journal

Ladder analytics

- Completion rate and median attempts to finish.
- Failure rate by step and strategy version.
- Largest position, required margin and capital-at-risk path.
- Expected drawdown and probability of losing starting capital.
- Comparison against fixed-risk sizing and no-ladder baseline.
- Paper results versus backtest assumptions versus manually imported real results.

Example

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Strategy: Gold Swing v2
Campaign: 10-step controlled ladder
Starting balance: $100
Risk per attempt: 1%
Maximum attempts per step: 3
Advance: target reached
Failure: step loss limit reached
Campaign stop: 10% peak-to-trough drawdown

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8. Deterministic backtesting and validation

The backtest engine, not the AI model, executes the strategy. It saves inputs, rules, dataset identity, assumptions, trades and metrics so every result is reproducible.

Protection	Purpose
Completed-candle execution	Avoid trading on information unavailable at decision time
No look-ahead	Prevent future values, revised data or future swing points from leaking
Costs	Apply spread, commission, slippage and financing where relevant
Futures handling	Record contract, roll rule, settlement and session boundaries
Out-of-sample	Separate strategy development from evaluation
Walk-forward	Test stability across changing market periods
Minimum sample	Warn when too few independent trades support a conclusion
Version lock	Prevent rules from changing inside an existing run

Core reports

- Net return, CAGR where meaningful, maximum drawdown, win rate and profit factor.
- Expectancy, average R, payoff ratio, Sharpe, Sortino and exposure.
- Consecutive wins/losses, margin demand and risk-of-ruin estimate.
- Equity curve, underwater curve, trade list and entry/exit chart markers.
- Performance by year, month, day, session, direction, setup and market regime.
- Monte Carlo ordering/resampling for ladder and progression risk.

Acceptance rule

A strategy cannot be marked validated merely because its optimized in-sample return is positive. It must pass data-quality, cost, sample-size and out-of-sample checks.

9. AI Strategy Copilot

AI is an interface to the formal system. It helps author, inspect and explain rules; deterministic services remain the source of market facts and test outcomes.

AI may	AI may not
Turn a plain-language idea into a draft strategy	Declare an unexecuted visual inspection a backtest
Ask for measurable definitions of vague terms	Change rules silently during a run
Suggest controlled parameter variations	Use future candles or revised values
Compare saved strategy versions and results	Invent market data, news, events or citations
Explain losing regimes and trade clusters	Override safety gates or connector quality states
Generate a readable test report	Present correlation as guaranteed causation

Conversation flow

USER IDEA -> AI DRAFT -> AMBIGUITY CHECK -> USER APPROVAL -> RULE SCHEMA
-> DETERMINISTIC BACKTEST -> SAVED TRADES/METRICS -> AI EXPLANATION

The AI response will identify provider/model, evidence snapshot and any missing inputs. When AI is unavailable, the deterministic backtest and Decision Desk continue to operate; only narrative assistance is reduced.

10. Three-pillar Decision Desk

Pillar	Gold-first evidence
Fundamental	DXY/broad dollar, nominal and real yields, inflation, Fed context, economic events, CFTC positioning and news risk
Technical	D1/H4 regime, H1 structure, M15 setup, M5 trigger, zones, ATR, RSI, MACD, VWAP and session context
Volume/Wyckoff	COMEX volume/open interest, relative volume, effort/result, profile, range phase and spring/test/upthrust/SOS/SOW evidence

Controlled states

State	Meaning
NO TRADE	Evidence is stale, unavailable, contradictory, event-blocked or insufficient
WATCH LONG	Bullish context exists; required trigger or confirmation is missing
WATCH SHORT	Bearish context exists; required trigger or confirmation is missing
LONG SETUP CONFIRMED	Alignment, volume, trigger, invalidation, event and risk gates pass
SHORT SETUP CONFIRMED	Alignment, volume, trigger, invalidation, event and risk gates pass

Minimum decision gates

- Fresh required data and declared source quality.
- At least two pillars align; volume must not oppose the direction.
- Exact trigger and invalidation are present.
- Configured reward-to-risk and event-risk rules pass.
- Manual checklist passes before a paper or external trade is linked.

Decision record

Every decision stores the exact data snapshot, pillar scores, sources, timestamps, trigger, invalidation, blockers and AI narrative. It can never be rewritten by later market data.

11. Journal and learning loop

Decisions and trades are separate records linked when appropriate. This preserves no-trade and missed-trade learning while keeping actual execution facts clean.

Decision record	Trade record
Three-pillar snapshot	Instrument, direction and external ticket/reference
Sources, quality and timestamps	Planned and actual entry/exit/stop/target
State, confidence and blockers	Size, risk, fees, slippage, P/L and R
Trigger and invalidation	MFE, MAE, session and timeframe
Strategy and ladder context	Screenshots, checklist and decision link
AI explanation	Emotions, mistakes, lessons and tags

Input methods

- Manual trade entry for external broker executions.
- CSV import with column mapping, preview, validation and duplicate protection.
- Screenshot attachments stored privately with user-scoped access.
- Missed-trade and no-trade records for discipline analysis.

Review outputs

Win rate, expectancy, average R, drawdown, rule adherence, strategy/setup performance, session performance, planned-versus-actual execution, psychology tags, missed-trade patterns, and ladder completion behavior.

12. Phased execution and acceptance gates

Phase	Deliverable	Exit gate
0. Blueprint and migration	Finalize screen map, schema, data contracts and old-feature mapping	Approved scope; no production mutation
1. Connector foundation	Secure Connector Hub, Vault, capability routing, health and audit	Keys masked; ownership and RLS tests pass
2. Data and charts	Databento, Twelve Data, FRED, CFTC, calendar/news, charting	Freshness/source shown; missing volume never inferred
3. Strategy Lab	General builder, versions and initial templates	Rules serialize, validate and remain immutable per run
4. Backtesting	Costs, multi-timeframe execution, metrics, walk-forward and Monte Carlo	Golden tests reproduce fills and metrics
5. AI Copilot	Natural-language drafting, ambiguity review and explanations	AI cannot bypass rule schema or fabricate results
6. Decision Desk	Three pillars, five states and saved evidence	Fail-safe gates and stale-data tests pass
7. Journal and ladder	Imports, attachments, psychology, campaign simulation	Migration reconciliation and duplicate tests pass
8. Alerts/order flow	TradingView webhooks, streaming, depth and heatmap	Reliability, licensing and rate controls verified

Release strategy

- Develop behind a protected route or feature flag.
- Use migrations with rollback-safe additions; do not overwrite historical records.
- Run unit, type, build, integration, RLS and live provider smoke tests.
- Test production-like data before navigation cutover.
- Retain an immediate rollback path during initial release.

13. Core records, governance and risks

Planned core records

Record	Purpose
connector_definitions	Supported providers, auth type and declared capabilities
user_connectors	User-owned connector metadata and encrypted-secret reference
connector_health_events	Connection tests, errors, latency and audit history
market_snapshots	Normalized, immutable evidence used by decisions/tests
strategies / strategy_versions	Editable strategy identity and immutable versioned rules
backtest_runs / backtest_trades	Reproducible configuration, metrics and fills
trading_decisions	Three-pillar state, gates, sources and narrative
ladder_campaigns / ladder_steps	Progression configuration and outcomes
trading_entries	Migrated and expanded manual/imported trade records
trade_attachments / import_batches	Private screenshots and controlled CSV ingestion

Principal risks and controls

Risk	Control
Credential exposure	Server-only decryption, masking, audit, rotation and least privilege
False volume conclusions	Declared source/proxy, null-safe calculations and no-trade gating
Overfitting	Out-of-sample, walk-forward, costs, minimum sample and version locks
Progressive-sizing ruin	Caps, margin model, drawdown stop, Monte Carlo and baseline comparison
AI hallucination	Tool/evidence grounding, deterministic engines and explicit missing-data states
Vendor dependency	Capability adapters, provider priority and fallback routing
Production regression	Feature flag, CI, security tests, smoke tests and rollback path

14. Initial execution checklist and references

Immediate implementation sequence

Step	Action
1	Reconcile the repository state and remove/rework uncommitted prototype files against this blueprint.
2	Create the protected Trading Lab route and feature boundary without changing the current production navigation.
3	Define connector capability contracts and secure metadata schema.
4	Implement server-only encrypted connector credential lifecycle and tests.
5	Add Databento/Twelve Data/FRED/CFTC adapters and normalized market-data contracts.
6	Implement the versioned strategy schema and validation layer.
7	Build deterministic backtest fixtures before adding AI authoring.

Definition of done

A phase is complete only when its automated checks, security controls, data-quality behavior and user-visible failure states are verified. Deployment alone is not completion.

Reference sources

Source	Reference
TradingView chart/datafeed boundaries	tradingview.com/charting-library-docs/latest/getting_started/product-comparison/
TradingView webhook alerts	tradingview.com/support/solutions/43000529348-how-to-configure-webhook-alerts/
Databento CME Globex data	databento.com/docs/knowledge-base/datasets
CME real-time futures/options API	cmegroup.com/market-data/real-time-futures-and-options-data-api.html
Bookmap API overview	bookmap.com/faq/
FRED series observations API	fred.stlouisfed.org/docs/api/fred/series_observations.html
CFTC COT explanatory notes	cftc.gov/MarketReports/CommitmentsofTraders/ExplanatoryNotes/index.htm
Supabase Vault	supabase.com/docs/guides/database/vault
Supabase API key security	supabase.com/docs/guides/getting-started/api-keys
TraderSync feature reference	tradersync.com/features/
Edgewonk psychology journal	edgewonk.com/trading-psychology

This blueprint is the agreed execution reference for the Buddies OS Trading Lab. Material scope changes should be recorded as a new version before implementation continues.